

PERCENTAGES — Short Revision Notes

Create Level: Produce Something New

1 Creating Profit and Loss Problems

Template 1: Create Profit Problem

Step	Action	Example
1	Choose Cost Price (CP)	Rs 500
2	Choose Profit %	20%
3	Calculate Profit Amount	$500 \times 20\% = 100$
4	Calculate Selling Price	$500 + 100 = 600$
5	Write the Problem	"A vendor buys an item for Rs 500 and sells it at a profit of 20%. What is the selling price?"

Create Your Own Profit Problem:

CP = _____

Profit% = _____

Profit Amount = _____

SP = _____

Problem: _____

Template 2: Create Loss Problem

Step	Action	Example
1	Choose Cost Price (CP)	Rs 800
2	Choose Loss %	15%
3	Calculate Loss Amount	$800 \times 15\% = 120$
4	Calculate Selling Price	$800 - 120 = 680$
5	Write the Problem	"A vendor buys an item for Rs 800 and sells it at a loss of 15%. What is the selling price?"

Create Your Own Loss Problem:

CP = _____

Loss% = _____

Loss Amount = _____

SP = _____

Problem: _____

2 Creating "Find CP" Problems

Template: Find CP from SP and Profit%

Step	Action	Example
1	Choose Cost Price (CP)	Rs 600
2	Choose Profit %	10%
3	Calculate Selling Price	$600 \times 110/100 = 660$
4	Write the Problem	"If an item is sold for Rs 660 at a profit of 10%, what was the cost price?"

Create Your Own:

SP = _____

Profit% = _____

CP = _____

Problem: _____

Template: Find CP from SP and Loss%

Step	Action	Example
1	Choose Cost Price (CP)	Rs 800
2	Choose Loss %	20%
3	Calculate Selling Price	$800 \times 80/100 = 640$
4	Write the Problem	"If an item is sold for Rs 640 at a loss of 20%, what was the cost price?"

Create Your Own:

SP = _____

Loss% = _____

CP = _____

Problem: _____

3 Creating Discount Problems

Template 1: Find SP with Discount

Step	Action	Example
1	Choose Marked Price (MP)	Rs 1500
2	Choose Discount %	10%
3	Calculate Discount Amount	$1500 \times 10\% = 150$
4	Calculate Selling Price	$1500 - 150 = 1350$
5	Write the Problem	"A shop offers a 10% discount on a shirt marked at Rs 1500. What is the selling price?"

Create Your Own Discount Problem:

MP = _____

Discount% = _____

Discount Amount = _____

SP = _____

Problem: _____

Template 2: Find Discount %

Step	Action	Example
1	Choose Marked Price (MP)	Rs 1000
2	Choose Discount Amount	Rs 200
3	Calculate Discount %	$200/1000 \times 100 = 20\%$
4	Write the Problem	"A shop reduces the price of an item from Rs 1000 to Rs 800. What is the discount percentage?"

Create Your Own:

MP = _____

SP = _____

Discount Amount = _____

Discount% = _____

Problem: _____

4 Creating Commission Problems

Template 1: Find Commission

Step	Action	Example
1	Choose Transaction Value	Rs 1,000,000
2	Choose Commission %	5%
3	Calculate Commission	$1,000,000 \times 5\% = 50,000$
4	Calculate Net Amount	$1,000,000 - 50,000 = 950,000$
5	Write the Problem	"A broker charges 5% commission on the sale of a house worth Rs 1,000,000. How much commission does the broker receive?"

Create Your Own Commission Problem:

Value = _____

Commission% = _____

Commission = _____

Net Amount = _____

Problem: _____

Template 2: Find Commission %

Step	Action	Example
1	Choose Transaction Value	Rs 500,000
2	Choose Commission Amount	Rs 25,000
3	Calculate Commission %	$25,000/500,000 \times 100 = 5\%$
4	Write the Problem	"A broker received Rs 25,000 commission on the sale of land worth Rs 500,000. What is the commission percentage?"

Create Your Own:

Value = _____

Commission = _____

Commission% = _____

Problem: _____

5 Creating Comparative Shopping Problems

Template: Compare Two Discounts

Step	Action	Example
1	Choose Marked Price	Rs 1500
2	Choose Shop A Discount	8%
3	Choose Shop B Discount	Rs 100 off
4	Calculate Shop A SP	$1500 \times 92\% = 1380$
5	Calculate Shop B SP	$1500 - 100 = 1400$
6	Write the Problem	"Shop A offers 8% discount and Shop B offers Rs 100 off on a Rs 1500 pair of shoes. Which is a better deal?"

Create Your Own Comparison Problem:

MP = _____

Shop A: _____

Shop B: _____

Shop A SP = _____

Shop B SP = _____

Better Deal = _____

Problem: _____

6 Creating Multi-Step Word Problems

Template: Mixed Selling Prices

Step	Action	Example
1	Choose item and quantities	60 kg manioc
2	Choose purchase price	Rs 50/kg
3	Calculate total CP	$60 \times 50 = 3000$
4	Choose different selling prices	20 kg at 70, 15 kg at 60, 10 kg at 50, 10 kg at 40, 5 kg discarded
5	Calculate total SP	$20 \times 70 + 15 \times 60 + 10 \times 50 + 10 \times 40 = 2900$
6	Find profit/loss	$3000 - 2900 = 100$ Loss

Create Your Own Mixed Selling Problem:

Item: _____

Quantity: _____

CP per unit: _____

Total CP: _____

Selling: _____

Problem: _____

7 Creating Markup and Discount Problems

Template: Markup + Discount

Step	Action	Example
1	Choose Cost Price	Rs 8000
2	Choose Markup %	25%
3	Calculate Marked Price	$8000 \times 125/100 = 10,000$
4	Choose Discount %	10%
5	Calculate Selling Price	$10,000 \times 90/100 = 9000$
6	Calculate Actual Profit	$9000 - 8000 = 1000$
7	Calculate Actual Profit %	$1000/8000 \times 100 = 12.5\%$

Create Your Own Markup + Discount Problem:

CP = _____

Markup% = _____

MP = _____

Discount% = _____

SP = _____

Actual Profit = _____

Actual Profit% = _____

Problem: _____

8 Creating Real-World Business Scenarios

Scenario 1: Fruit Vendor

Part	Details	Your Creation
Product	Mangoes	_____
Quantity	100 fruits	_____
Buying Price	Rs 18 each	_____
Spoiled	20 fruits	_____
Selling Price	Rs 30 each (for remaining)	_____
Question	Find profit/loss and profit%	_____

Create Your Own:

I bought _____ for Rs _____ each.

_____ were spoilt.

I sold the remaining at Rs _____ each.

Find my profit/loss.

Scenario 2: Multiple Sellers

Part	Details	Your Creation
First Seller	Carpenter makes chair for Rs 1800	_____
First Profit	20%	_____
Second Seller	Vendor buys from carpenter	_____
Second Profit	20%	_____
Question	Find final selling price and profit%	_____

Create Your Own:

Person A makes _____ for Rs _____.

Person A sells to Person B at _____% profit.

Person B sells to customer at _____% profit.

Find the final price.

Scenario 3: Discount and Commission

Part	Details	Your Creation
Item	House	_____
Price	Rs 5,000,000	_____
Discount	1% to buyer	_____
Commission	3% to broker	_____
Question	Find seller's final income	_____

Create Your Own:

A person sells _____ for Rs _____.

He gives _____% discount to buyer.

He pays _____% commission to broker.

How much does he receive?

9 Creating Your Own Worksheet

Worksheet: Profit and Loss

Problem	Your Solution
CP=400, SP=500. Find Profit%	_____
CP=800, Loss=20%. Find SP	_____
SP=660, Profit=10%. Find CP	_____
SP=720, Loss=10%. Find CP	_____

Create Your Own Worksheet

Your Problem	Answer Space
_____	_____
_____	_____
_____	_____
_____	_____

10 Creating Your Own Answer Key

Example Answer Key

Problem	Answer	Steps
CP=500, SP=600. Find Profit%	20%	$100/500 \times 100 = 20\%$
CP=800, SP=680. Find Loss%	15%	$120/800 \times 100 = 15\%$
CP=600, Profit=25%. Find SP	750	$600 \times 125/100 = 750$
MP=1000, Disc=20%. Find SP	800	$1000 \times 80/100 = 800$

Create Your Own Answer Key

Your Problem	Your Answer	Your Steps
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

11 Creating Percentage Conversion Table

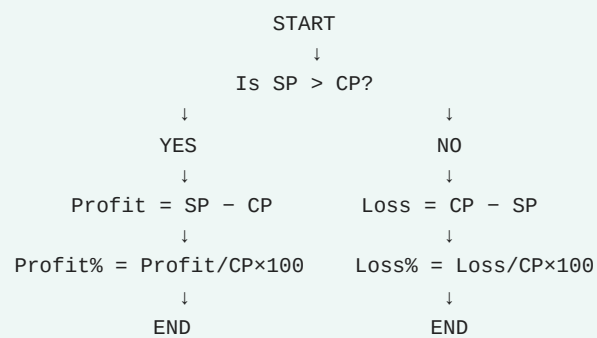
Percentage	Fraction	Decimal
10%	1/10	0.1
20%	1/5	0.2
25%	1/4	0.25
50%	1/2	0.5
75%	3/4	0.75
100%	1	1.0

Complete Your Own Table

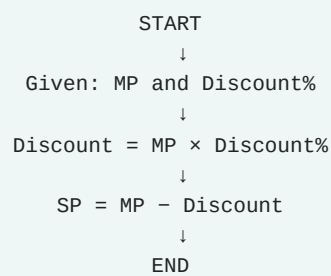
Percentage	Fraction	Decimal
5%	_____	_____
30%	_____	_____
40%	_____	_____
60%	_____	_____
80%	_____	_____

12 Creating Flowcharts

Flowchart: How to Solve Profit/Loss Problems



Flowchart: How to Solve Discount Problems



Create Your Own Flowchart:

Topic: _____

Flowchart:

_____ → _____ → _____ → _____

13 Creating Step-by-Step Solutions

Template: Solving a Percentage Problem

Step	Action	Example
Step 1	Identify what is given	CP=800, SP=1000
Step 2	Identify what to find	Profit%
Step 3	Find Profit/Loss	Profit=200
Step 4	Apply formula	$200/800 \times 100$
Step 5	Calculate answer	25%

Create Your Own Step-by-Step:

Problem: _____

Step 1: _____

Step 2: _____

Step 3: _____

Step 4: _____

Step 5: _____

Answer: _____

14 Creating Problem-Solving Templates

Template 1: Fill in the Blanks

A vendor buys _____ items at Rs _____ each.
He sells them at Rs _____ each.
His total CP = _____ $\times$ _____ = Rs _____
His total SP = _____ $\times$ _____ = Rs _____
His Profit/Loss = _____ - _____ = Rs _____
His Profit/Loss% = _____ / _____ $\times$ 100 = _____%

Template 2: Fill in the Blanks (Discount)

An item is marked at Rs _____.
A discount of _____% is offered.
Discount Amount = _____ $\times$ _____% = Rs _____
Selling Price = _____ - _____ = Rs _____

Template 3: Fill in the Blanks (Commission)

A broker sells _____ worth Rs _____.
Commission = _____%
Commission Amount = _____ $\times$ _____% = Rs _____
Seller Receives = _____ - _____ = Rs _____

15 Creating Checklists

Profit/Loss Checklist

- ☒ PROFIT/LOSS CHECKLIST:
- ☐ Did I find the correct profit/loss amount?
 - ☐ Did I use the correct base (CP)?
 - ☐ Did I multiply by 100?
 - ☐ Did I add the % sign?
 - ☐ Is my answer reasonable?
 - ☐ $SP > CP \rightarrow \text{Profit}$
 - ☐ $SP < CP \rightarrow \text{Loss}$
 - ☐ $SP = CP \rightarrow \text{Break-even}$

Discount Checklist

- ☒ DISCOUNT CHECKLIST:
- ☐ Did I use MP as base?
 - ☐ Did I calculate discount correctly?
 - ☐ Did I subtract discount from MP?
 - ☐ Did I add the % sign?
 - ☐ Is my answer reasonable?
 - ☐ $\text{Discount}\% = (\text{Discount}/\text{MP}) \times 100$
 - ☐ $SP = MP - \text{Discount}$

16 Self-Test — Creation Questions

No.	Task	Your Answer
1	Create a profit problem with CP=600, Profit=20%	_____
2	Create a loss problem with CP=900, Loss=10%	_____
3	Create a discount problem with MP=2000, Discount=15%	_____
4	Create a commission problem with Value=500,000, Commission=5%	_____
5	Create a comparison shopping problem	_____
6	Create a word problem involving profit and loss	_____
7	Create an answer key for 3 percentage problems	_____
8	Create a flowchart for solving discount problems	_____
9	Create a fill-in-the-blanks problem	_____
10	Create a markup + discount problem	_____

17 Creation Templates Summary

CREATE YOUR OWN:

1. PROFIT PROBLEM: $CP + \text{Profit\%} \rightarrow SP$
2. LOSS PROBLEM: $CP + \text{Loss\%} \rightarrow SP$
3. FIND CP PROBLEM: $SP + P/L\% \rightarrow CP$
4. DISCOUNT PROBLEM: $MP + D\% \rightarrow SP$
5. COMMISSION PROBLEM: $\text{Value} + C\% \rightarrow \text{Commission}$
6. COMPARISON PROBLEM: Two deals $\rightarrow$ Which is better?
7. MIXED SELLING: Different prices $\rightarrow$ Total P/L
8. MARKUP + DISCOUNT: $CP \rightarrow MP \rightarrow SP \rightarrow \text{Actual P/L}$
9. WORD PROBLEM: Real-world context
10. WORKSHEET: Multiple problems with answer key

Quick Creation Guide

PROFIT PROBLEM CREATION:

CP = Choose a number
Profit% = Choose a percentage (10%, 20%, 25%, 50%)
 $SP = CP \times (100 + \text{Profit\%})/100$

LOSS PROBLEM CREATION:

CP = Choose a number
Loss% = Choose a percentage (10%, 15%, 20%, 25%)
 $SP = CP \times (100 - \text{Loss\%})/100$

DISCOUNT PROBLEM CREATION:

MP = Choose a number
Discount% = Choose a percentage (10%, 15%, 20%, 25%)
 $SP = MP \times (100 - \text{Discount\%})/100$

COMMISSION PROBLEM CREATION:

Value = Choose a number
Commission% = Choose a percentage (2%, 3%, 5%, 10%)
Commission = $\text{Value} \times \text{Commission\%}/100$
Net = $\text{Value} - \text{Commission}$

COMPARISON PROBLEM CREATION:

Choose MP
Choose two discount types (e.g., 10% vs Rs 100 off)
Calculate both SPs
Compare and justify which is better

Revision Tip: Creating your own problems is the best way to master percentages! Start with simple problems, then make them more complex. Always solve your own problems to check they make sense. The ability to create problems shows true understanding!